

DevOps & CI/CD

Culture, practices and automation connecting development, operations, quality and security through continuous feedback.

What is DevOps?

DevOps combines culture, collaboration, processes and technology across the software lifecycle. It aims to reduce silos and handoffs, accelerate feedback and make delivery and operations more reliable. DevOps is not a tool, job title or synonym for a pipeline.

Version control and collaboration

Version control makes changes traceable and supports collaboration, review and automation. Code, configuration and, where appropriate, infrastructure can be managed as versioned assets.

Continuous Integration — CI

CI integrates changes frequently and runs automated builds and checks to detect problems early. A good CI pipeline provides fast, reliable feedback; test volume without reliability creates noise.

Continuous Delivery and Deployment

Continuous Delivery keeps software in a releasable state through automated build, tests and release preparation. Continuous Deployment goes further by automatically promoting approved changes to production when criteria are met.

Infrastructure as Code and environments

IaC describes infrastructure and configuration in repeatable, versionable form. Consistent environments reduce “works on my machine” issues and improve defect reproduction.

Observability and monitoring

Logs, metrics and traces help understand production behavior. Monitoring availability, errors, latency and business signals closes the feedback loop and enables learning from real usage.

DevSecOps

Security can be integrated with dependency analysis, secret scanning, SAST/DAST and automated policies instead of being a final audit.

Where QA fits in DevOps

QA contributes from planning and testability through multi-layer automation, meaningful quality gates, post-deploy validation, observability and incident learning. The goal is useful feedback, not simply putting every test in the pipeline.



Watch out

Using Jenkins, Azure DevOps, Docker or GitHub Actions does not mean a team practices DevOps. Tools cannot fix silos, slow feedback or unreliable pipelines on their own.